$$\begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix} \xrightarrow{E_{21}} \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix} = U$$

$$\left(E_{21}^{-1}\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\right)\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$L \quad D \quad U$

$y = b_2$

$x + 2y = b_1$

solve for x:

$$\underbrace{\begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix}}_{A}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 15 \end{bmatrix}$$

$Ax = b$

$LDUx = b$

Ux → vector $= y$

$LDy = b$

$y = ?$

$Ux = ?$

Inverse:

$$\left[\begin{array}{cc|cc} 2 & 4 & 1 & 0 \\ 4 & 11 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{cc|cc} 2 & 4 & 1 & 0 \\ 0 & 3 & -2 & 1 \end{array}\right] = \left[\begin{array}{cc|cc} 1 & 2 & 1/2 & 0 \\ 0 & 1 & -2/3 & 1/3 \end{array}\right]$$

$$= \left[\begin{array}{cc|cc} 1 & 0 & 11/6 & -2/3 \\ 0 & 1 & -2/3 & 1/3 \end{array}\right]$$

A^{-1}

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x+2y \\ y \end{bmatrix}$$

L A Ux = y

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2x+4y \\ 3y \end{bmatrix} \quad / : \; 4x+8y + 3y$$

$$\begin{bmatrix} 2x+4y \\ 4x+11y \end{bmatrix} = \begin{bmatrix} 6 \\ 15 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix}$$

Start

Vector space

Condition

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

A b

↳ not solvable for all right hand side. must be in my col space. → if not (vector is

$$x\begin{bmatrix}1\\2\\3\\4\end{bmatrix} + y\begin{bmatrix}1\\1\\1\\1\end{bmatrix} + z\begin{bmatrix}2\\3\\4\\5\end{bmatrix} = \begin{bmatrix}b_1\\b_2\\b_3\\b_4\end{bmatrix}$$

outside of my col space
→ not solvable

col space = C(A)
↓
taking linear combinations of the columns

→ 4 rows (each col has 4 components)

C(A) = Plane in $\mathbb{R}^4$
↓
2 Linearly independent.

→ boring sol^n.

$$\underbrace{\begin{bmatrix}1&1&2\\2&1&3\\3&1&4\\4&1&5\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}0\\0\\0\end{bmatrix}}_{x} = \underbrace{\begin{bmatrix}0\\\\\end{bmatrix}}_{b}$$

$k\begin{bmatrix}1\\1\\-1\end{bmatrix}$ → fun sol^n

$Ax = 0$

Set: { of All sol^n $k\begin{bmatrix}1\\1\\-1\end{bmatrix}$ } → A line

↓
closed for multiplication
↓
Add another vector can't escape the line

so both cond^n satisfied. IT IS A SPACE

↳ NULL SPACE!

set of all vectors which after tranformation of of matrix becomes 0? Solves $Ax = 0$

After transforming all the vectors in the matrix collapse into a single point or NULL space.

#
$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$
(x2, x2)

find col space & Row space.

=> Elimination

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$
(x2, x2)

ignore

move on

$$\therefore \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

upper ~~triangle~~

echelon (stairs) (level)

2 independent columns.

of pivots = 2 = rank

1st & 3rd col => pivot col

others → free col

% $$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$ need 1 along the pivots.

$$\rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Reduced Row echelon form.

rref(A)